

Purification of nucleic acids

Risk assessment

- Guanidine salts form TOXIC GASES with bleach or strong acids
- ▷ Wear gloves, safety glasses, lab coat
- DO NOT add bleach or acidic solutions directly into sample waste



Reviewed: Jun 10, 2023

This is a bench card. Full protocol available online.

A > Purification by alcohol precipitation

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| ☐  R0045 3 M Sodium acetate | ☐ Buffer PE (Wash buffer) |
| ☐ Ethanol, ice-cold | ☐  R0056 5 mM Tris hydrochloride |

- (1.) Mix 1.0 vol sample with 0.1 vol 3 M sodium acetate. Mix well.
 - (2.) *Optional:* Add 10–20 µg/mL linear polyacrylamide to precipitate picogram quantities of nucleic acids.
 - (3.) Add 2.0 vol ethanol to precipitate DNA, or 2.5–3.0 vol ethanol to precipitate RNA. Mix by vortexing.
 - (4.) Precipitate for 15–30 min on ice.
 - (5.) Centrifuge at 12 000 × *g* for 15 min at 4 °C. Discard supernatant and spin briefly to remove residual solvent. The pellet will appear nearly transparent.
 - (6.) Wash pellet with 650 µL Buffer PE or freshly prepared 70% ethanol. Discard supernatant carefully.
 - (7.) Dry pellet for 10–15 min at room temperature, or for 1 min in a vacuum centrifuge.
- Critical:** Avoid overdrying, which impairs dissolution and may denature small dsDNA.
- (8.) Resuspend in 15–30 µL 5 mM Tris pH 8.0 or buffer of choice.

B > **Purification over a silica matrix**

- ☐ Buffer PB (Binding buffer)
- ☐ Buffer PE (Wash buffer)

☐  R0056 5 mM Tris hydrochloride

- (1.) Mix 1.0 vol sample with 2.0 vol Buffer PB. 
- (2.) Apply up to 700 μL to a silica spin column. Centrifuge at $11\,000 \times g$ for 30 s or use vacuum manifold 
- (3.) Wash with 650 μL Buffer PE.

Critical: Let stand for 2–5 min before spinning if DNA will be used in salt-sensitive applications such as blunt-end ligation. 

- (4.) Repeat wash to minimize salt carryover.
- (5.) Dry membrane by centrifugation at $11\,000 \times g$ for 1 min.

Critical: Residual ethanol may persist if flow-through is not discarded prior to drying. 

- (6.) Place spin column in a clean tube. Apply 15–30 μL 5 mM Tris pH 8.0 onto the silica membrane. Incubate 1 min. Elute at $11\,000 \times g$ for 1 min.  
- (7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

 Recipe (available online)  Troubleshooting (available online)  Notes (available online)

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